

Question number	Answer	Mark
6(a)	The only correct answer is C <i>A is incorrect because only statement 1 is incorrect as proteins contain oxygen as well</i> <i>B is incorrect because only statement 1 is incorrect as proteins contain oxygen as well</i> <i>D is incorrect because only statement 1 is incorrect as proteins contain oxygen as well</i>	(1)

Question number	Answer	Mark
6(b)	The only correct answer is D <i>A is incorrect because there are carbon – carbon double bonds in an unsaturated fatty acid</i> <i>B is incorrect because there are carbon – carbon double bonds in an unsaturated fatty acid</i> <i>C is incorrect because the ratio is higher ie fewer hydrogens for the same number of carbons</i>	(1)

Question number	Answer	Additional guidance	Mark
6(c)	haemoglobin (1)	Examiners will need to check any other answers ACCEPT myoglobin, leghaemoglobin, neuroglobin, transferrin, ferritin, ferredoxin, cytochrome, catalase, peroxidase IGNORE Hb haem group	(1)

Question number	Answer	Additional guidance	Mark
6(d)	An explanation that makes reference to the following points: • because antioxidants reduce free radicals (1) • therefore {cell damage / damage to lining of blood vessels / oxidative stress} will be {reduced / prevented} (1) • (less cell damage / antioxidants) {reduce/ prevent} {plaque / atheroma} formation (due to decreased free radicals) (1) • (antioxidants / less plaque) reduce {cardiovascular disease / CVD / heart disease / heart attack / atherosclerosis / stroke} (1)	NB ACCEPT converse of mps 2, 3 and 4 i.e. what free radicals do ACCEPT neutralise / donate electrons to / break down / stabilise ACCEPT reduces cholesterol build up	(4)

Question number	Answer	Additional guidance	Mark
6(e)(i)	2.9 (1)	ACCEPT $\times 2.9$ / $2.9 \times$ / 2.9 times DO NOT ACCEPT 2.90 or any other value	(1)

Question number	Answer	Additional guidance	Mark
6(e)(ii)	180 / 1800 / 95 / 94.7 / 94.74 (%) (1)		(1)

Question number	Answer	Additional guidance	Mark
6(e)(iii)	An answer that makes reference to three of the following points: • different organisms contain different levels of antioxidant activity (1) • (overall) plant eating organisms have more antioxidant activity (than animal eating organisms) (1) • (overall) insects have more antioxidant activity (than arachnids) (1) • there is more variation in the antioxidant activity in plant eating organisms (than animal-eating organisms) (1)	ACCEPT converse Accept AO ACCEPT the error bars are larger in the plant eaters	(3)